



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000

WEEK 26

TUTOR GUIDE

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RELEVANT PRIOR LEARNING

MED 1000

- 1.07 – Viruses and bacteria [L]
- 1.31 – Bacterial pathogen in URT infections [L]
- 1.31 – History and examination of the ENT [L & CCS]

MED 2000

- 2.02 – Microbiology of common skin infections [L]
- 2.06 – History and examination of the ENT [CCS]
- 2.06 – Assessment of neck lumps [L]

WEEK 26_PAEDIATRICS_ENT_DERMATOLOGY		
Yr	Wk	LEARNING OBJECTIVES
4	26	• Revise the anatomy of the ear, in particular the anatomy of hearing and be able to explain the pathophysiology of acute suppurative otitis media and otitis media with effusion. • Know the key components of the assessment of the child with possible ear infection and/or hearing problems due to otitis media with effusion. • Know the risk factors that predispose children to middle ear infection. • Know the possible sequelae of acute suppurative otitis media and the role of medical management in reducing these sequelae. • Explain the pathophysiology of cholesteatoma. • Know the management of acute otitis media and otitis media with effusion. • Outline the natural history of otitis media with effusion and when referral for hearing assessment and surgical opinion is required. • Be able to interpret an audiogram and explain the findings to parents. • Outline the role of tympanostomy tubes [grommets] in the management of otitis media with effusion.
4	26	• Know the key features of the assessment of the child with an acute sore throat. • Know the clinical features that suggest a possible serious condition. • Know the key clinical features of acute tonsillitis. • Outline the clinical features which assist in differentiating between a viral and bacterial causative organism. • Be able to outline the pros and cons of the use of antibiotics in acute pharyngitis. • Detail the indications and contraindications for tonsillectomy and adenoidectomy.
4	26	• Know the key components of the assessment of the child presenting with a rash in the napkin area. • Know the key clinical features that assist in distinguishing between irritant dermatitis, candidiasis, seborrhoeic dermatitis, miliaria and psoriasis. • Be able to outline a management plan for irritant dermatitis.
4	26	• Know the key clinical features for the following rashes that present in children; impetigo, mollusca contagiosa, tinea, scabies, coxsackie B [hand, foot & mouth], herpes simplex and warts. • In a table outline the pathophysiology, typical presentation and management of each of the above rashes.

WEEK 26_PAEDIATRICS_ENT_DERMATOLOGY		
YR	WK	LEARNING OBJECTIVES
4	26	• Outline the key differences between haemangiomas and vascular malformations. • Outline the possible complications of haemangiomas presenting in children. • Know the key components of the management of haemangiomas. • Briefly outline the different types of vascular malformations and how they can present clinically.
4	26	• Outline a systematic approach to the assessment of midline and lateral neck swellings in children [SDL] • Outline the key clinical features of cervical lymphadenitis due to atypical mycobacterium infection in children. [SDL] • Explain branchial remnants and the variety of abnormalities that may occur.[SDL] • Briefly outline the causes of torticollis in children. [SDL]

CASE ONE

Short case number: 4_26_01

Category: Children and Young People; Respiratory & ENT system

Discipline: Paediatrics Surgery_ENT

Setting: General Practice

Topic: Otitis Media with effusion.

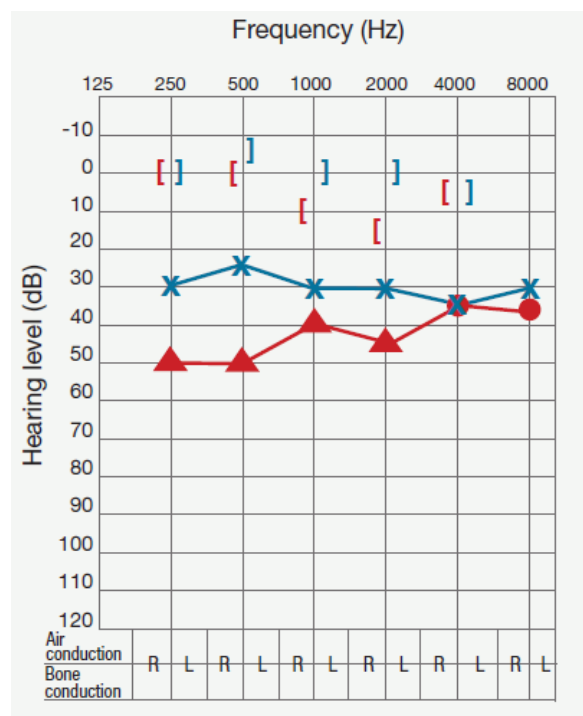
CASE



Dawn Munro is 3 years old, she is well known to the GP practice because of her frequent problems with middle ear infections. Dawn presents with her mother Elizabeth, who is concerned that Dawn's frequent ear infections are affecting her hearing. She would like you to check Dawn's ears again, to make sure there is no infection.

QUESTIONS

1. What would be the key components of your assessment of Dawn and why?
2. On examination you note that Dawn's language development is within normal range for her age, but on examination there are clinical features of bilateral otitis media with effusion. What are the key clinical features of otitis media with effusion?
3. Elizabeth asks you what causes the 'fluid to build up' in Dawn's ears. What would you explain to her?
4. What is the natural history of otitis media with effusion?



5. You decide to refer Dawn for an audiogram because of the concerns regarding her hearing. Dawn's audiogram result is shown here; explain what this audiogram demonstrates.
6. Dawn's audiogram results support a diagnosis of bilateral otitis media with effusion [glue ear]. What would be your next step in management and why?
7. Elizabeth comments that her niece needed grommets to fix her glue ear, 'does Dawn need to have an operation as well?' What would you explain to Elizabeth about the indications for tympanostomy tubes in this clinical situation? Outline the complications of this procedure.

Resources:

1. Benjamin B, Ear, nose and throat and head and neck surgery problems. Chapter 22 in Practical Paediatrics. Edition 6. Edited by Robertson DM & South M. Churchill Livingstone, Elsevier. [pages 791-798]
2. Harris C. Childhood ENT disorders. Australian Family Physician. Vol 31, No8, August 2002.
3. Coates H. How to interpret audiograms. Therapy Update, Australian Doctor. 25th May 2007.

ANSWERS

1. What would be the key components of your assessment of Dawn and why?

History

- Immunisations
- Family history of grommets, otitis media etc
- PMHx
- Recurrent acute otitis media
- Behaviour, hearing , milestone development etc

Examination

- Growth centiles
- Physical Milestones
- ENT examination.

2. On examination you note that Dawn's language development is within normal range for her age, but on examination there are clinical features of bilateral otitis media with effusion. What are the key clinical features of otitis media with effusion?

The clinical features of OME, common up to 8-10 years of age, are usually seen first in winter and are sometimes variable and unpredictable. Symptoms often follow a viral upper respiratory tract infection or incompletely resolved ASOM. Because at first there may be no symptoms the condition may remain unrecognized. Earache and deafness are two important features.

Earache presents in two ways. Firstly, as a flare-up during an acute respiratory infection there may be typical ASOM with severe pain. Secondly, repeated 'small' earaches, lasting for minutes rather than hours, that may wake the child at night or occur at school and settle quickly.

Recognition of physical signs in the tympanic membrane is often difficult for the inexperienced: a good pneumatic otoscope with magnification is invaluable. The common physical signs include:

- yellow or amber appearance of the tympanic membrane
- vascular dilatation, which is easier to see with magnification
- thickening and dullness of the tympanic membrane
- indrawing of the tympanic membrane, giving a concave appearance so that the handle of the malleus appears short
- in advanced cases there may be atelectasis in the middle ear with atrophy and thinning of the tympanic membrane

Sluggish or poor movement of the tympanic membrane detected using a pneumatic otoscope is a most important physical sign. The thicker the fluid, the less the 'bounce'. In children old enough to perform a pure tone audiogram a mild to moderate conduction deafness will be detected.

Impedance tympanometry tests the bounce of the tympanic membrane and typically shows a 'flat' curve.

3. Elizabeth asks you what causes the 'fluid to build up' in Dawn's ears. What would you explain to her?

The aetiology is uncertain but the ventilation, drainage and clearing mechanism of the eustachian tube is abnormal. Organisms similar to those found in ASOM can be cultured in 30-50% of cases - the effusion apparently follows incomplete resolution of ASOM. A mucoid, non-purulent effusion, containing leukocytes, dead or live bacteria, serum protein and mucus, accumulates in the middle ear. The effusion may be thin, thick, gelatinous or, in advanced cases, even 'rubbery'.

4. What is the natural history of otitis media with effusion?

Seventy percent of children still have an effusion after AOM after 2 weeks, 40% at one month, 20% at 2 months and 10% at 3 months. No intervention is needed up until this time. Children do not need to be referred for surgical intervention in the first three months following an acute infection, as spontaneous resolution will occur in 90%.

5. You decide to refer Dawn for an audiogram because of the concerns regarding her hearing. Dawn's audiogram result is shown here; explain what this audiogram demonstrates.

Conductive hearing loss with a larger hearing loss to the right.

Conductive hearing loss is the result of a blockage either in the outer or middle ear. The typical conductive hearing loss we see in children is usually the result of otitis media with effusion (OME) or glue ear. In adults it may be the result of otosclerosis or a tympanic membrane perforation. On the audiogram a conductive loss is shown with a difference, or gap, between the air and bone thresholds, which is called the air-bone gap.

6. Dawn's audiogram results support a diagnosis of bilateral otitis media with effusion [glue ear]. What would be your next step in management and why?

Children with effusions persisting more than three months who have hearing loss or associated symptoms (eg behavioural or developmental problems) should be referred for formal audiological assessment before referral. Benefits of tympanostomy tubes (grommets) to drain effusions include a marked improvement in hearing. However, this has been demonstrated to be of short term benefit only and needs to be weight up against the disadvantages of having ventilations tubes in situ.

7. Elizabeth comments that her niece needed grommets to fix her glue ear, 'does Dawn need to have an operation as well?' What would you explain to Elizabeth about the indications for tympanostomy tubes in this clinical situation? Outline the complications of this procedure.

This procedure is associated with the risks of a general anaesthetic and the sequelae of indwelling tubes including otorrhoea, obstruction, granulation tissue, premature extrusion, medial displacement. Long term sequelae include focal atrophy, chronic perforation, retraction pockets.

CASE TWO

Short case number: 4_26_02

Category: Respiratory & ENT System; Children & young people

Discipline: Paediatrics Surgery_ENT

Setting: General Practice

Topic: Acute Sore Throat_tonsillitis

CASE



Max Allcorn is 7 years old, he presents with a sore throat and temperature of 38.7°C. His mother is concerned she thinks he has tonsillitis again because his breath smells.

“this will be the third time this year.....do you think it is time to take his tonsils out?”

QUESTIONS

1. What are the key clinical features that support a diagnosis of tonsillitis?
2. On assessment of Max, you note the appearance of his tonsils [as pictured], he is febrile with tender cervical lymphadenopathy.
 - a) What key clinical features would concern you that a more serious condition was present?
 - b) What clinical features would require urgent transfer to hospital?
3. What clinical features may assist in distinguishing between a viral and bacterial causative organism?
4. Your diagnosis is acute bacterial tonsillitis, outline your management plan, including investigations, management of symptoms, choice of antimicrobial agent and follow-up plan.
5. When would you consider referral to an ENT specialist to discuss tonsillectomy?
6. Outline the current recommendations regarding the use of antibiotics in the management of acute sore throat and the risks of overuse of antibiotics.

Resources:

1. Benjamin B, Ear, nose and throat and head and neck surgery problems. Chapter 22 in Practical Paediatrics. Edition 6. Edited by Robertson DM & South M. Churchill Livingstone, Elsevier. [pages 800-803]
2. Harris C. Childhood ENT disorders. Australian Family Physician. Vol 31, No8, August 2002.
3. Therapeutic Guidelines 2010 – Antibiotics

ANSWERS

1. What are the key clinical features that support a diagnosis of tonsillitis?

Acute tonsillitis has been defined clinically as a condition in which not only is inflammation mostly confined to the tonsils but the clinical features also include acute sore throat, fever, difficulty in swallowing, enlarged tender regional cervical lymph nodes, halitosis and constitutional symptoms such as lethargy, nausea and vomiting. Sometimes there is abdominal pain. Examination can show various appearances in tonsillitis

- red mucosa over the tonsils and oedematous, generalized inflammation (parenchymatous tonsillitis)
- yellowish-white exudate in the crypts of the tonsils (follicular tonsillitis)
- the crypts become filled with 'debris', an exudate of desquamated epithelium and pus
- coalescence of these follicles can form a thin, white, non-confluent, patchy membrane that peels away without bleeding (membranous tonsillitis)
- the typical redness, oedema and purulent secretion (exudative tonsillitis). Note that this appearance is not necessarily diagnostic of streptococcal infection

2. On assessment of Max, you note the appearance of his tonsils [as pictured], he is febrile with tender cervical lymphadenopathy.

- a) What key clinical features would concern you that a more serious condition was present?
- b) What clinical features would require urgent transfer to hospital?

In a small number of cases acute sore throat can be a medical emergency. Infectious mononucleosis, peritonsillar cellulitis, quinsy or severe tonsillitis can result in airway obstruction requiring urgent hospital referral. Epiglottitis should also be considered as a differential diagnosis. If breathing difficulty is present, attempts to examine the throat should be avoided.

Symptoms that suggest a sore throat is a medical emergency requiring urgent transfer to hospital:

- Stridor
- Respiratory distress
- Drooling
- Inability to swallow
- Trismus (inability to open mouth fully)

3. What clinical features may assist in distinguishing between a viral and bacterial causative organism?

It is difficult to distinguish clinically between viral and bacterial causes of sore throat. In particular, Epstein–Barr virus infection ('glandular fever', infectious mononucleosis) is easily confused with bacterial causes.

Of the bacterial causes, *Streptococcus pyogenes* is the organism of greatest concern. There are four diagnostic features that increase the probability that *S. pyogenes* is the cause of infection—

fever greater than 38 °C, tender cervical lymphadenopathy, tonsillar exudate, and no cough. Children aged 3 to 14 years are also more likely to have streptococcal infection (30% to 40% of sore throats are caused by *S. pyogenes* in this age group, compared to 5% to 15% in older patients, and it is even rarer in younger children).

4. Your diagnosis is acute bacterial tonsillitis, outline your management plan, including investigations, management of symptoms, choice of antimicrobial agent and follow-up plan.

The difficult dilemma is that, although it might often seem reasonable to give antibiotics to a sick child with a sore throat, available evidence does not support their administration whether the infection is streptococcal or viral. When laboratory confirmation of a bacterial cause is available it is rational to give antibiotics to those at risk of remote complications such as post-streptococcal rheumatic fever or, less likely, acute glomerulonephritis. In others, the risk of these complications is not increased by delay in giving antibiotics for 24-48 hours: in fact treatment begun within 7-9 days of the onset is effective. Bear in mind that 10-20% of otherwise normal children may be carriers of *S. pyogenes* and that considerably fewer than 50% of cases of AST are eventually proved to be bacterial. In many cases no pathogen is isolated.

Antibiotics confer minimal worthwhile benefit (symptoms may last for 1 day less) over purely symptomatic treatment (rest, paracetamol and fluids) and are seldom effective in preventing suppurative complications except in high-risk groups. Nevertheless, pending the result of a throat swab, it might be prudent to immediately treat those who are unduly ill or toxic or those at risk, including diabetics, immunocompromised individuals and Aboriginal, Torres Strait and Pacific Islander children

5. When would you consider referral to an ENT specialist to discuss tonsillectomy?

Recurrent sore throats (all criteria)

- Sore throats are due to tonsillitis
- Symptoms for at least a year
- At least five episodes in one year
- Episodes are disabling and prevent normal functioning
- Sustained or increasing morbidity

Chronic tonsillitis

Possibility of malignancy eg. lymphoma

Persistent/progressive asymmetrical enlargement

Quinsy

6. Outline the current recommendations regarding the use of antibiotics in the management of acute sore throat and the risks of overuse of antibiotics.

The therapeutic goals used to justify antibiotic treatment for pharyngitis/tonsillitis are:

- to prevent nonsuppurative complications (eg acute rheumatic fever, acute glomerulonephritis)
- to prevent suppurative complications (eg acute otitis media, quinsy, acute sinusitis)
- to shorten the illness.

Antibiotics *are* recommended to prevent nonsuppurative complications in the following high-risk populations:

- patients aged 2 to 25 years with sore throat in communities with a high incidence of acute rheumatic fever (eg some Indigenous communities in central and northern Australia, and some other underprivileged communities)
- patients of any age with existing rheumatic heart disease
- patients with scarlet fever.

Except in these high-risk populations, nonsuppurative complications are now so rare in Australia that they are no longer an indication for antibiotic use. For example, the annual incidence of rheumatic fever of less than 1 case per 100 000 is much less than the risk of a severe reaction to penicillin (10 to 40 per 100 000 treatment courses).

S. pyogenes remains highly susceptible to penicillin. When treatment is indicated, a suitable antibiotic regimen is phenoxymethylpenicillin 500 mg (child: 15 mg/kg up to 500 mg) orally, 12-hourly for 10 days.

CASE THREE

Short case number: 4_26_03

Category: Children & Young people; Musculoskeletal & Skin

Discipline: Paediatrics Medicine_Dermatology

Setting: General Practice

Topic: Napkin Rash in children.

CASE



Olivia Hanson is 10 weeks old, she presents with her mother who is concerned about Olivia's nappy rash, which improves initially with nappy rash creams but then comes back again. *"I have stopped using disposable nappies, but it does not seem to be getting better, it is making her very irritable and she does not seem to be settling as well"*

QUESTIONS

1. Why do babies commonly develop a rash in the nappy area and what is the most frequent type of nappy rash seen?
2. In assessing Olivia, what are the key components of your history and examination and why?
3. You note that Olivia, is alert, active and thriving, in assessing the nappy rash you note the clinical features are consistent with irritant dermatitis. What clinical features support this diagnosis? How would the clinical findings differ in candidiasis, seborrhoeic dermatitis, miliaria and psoriasis?
4. What management plan would you recommend to Olivia's mother?

Resources:

1. Rogers M. Skin disorders in infancy and childhood, Chapter 21 in Practical Paediatrics. Edition 6. Edited by Robertson DM & South M. Churchill Livingstone, Elsevier. [pages 776-777]
2. Royal Children's Hospital Melbourne Fact Sheets – Kids Health Info for Parents. Nappy Rash

ANSWERS

1. Why do babies commonly develop a rash in the nappy area and what is the most frequent type of nappy rash seen?

The most common cause of nappy rash (also called diaper dermatitis) in babies is irritation. Constant moisture and rubbing can cause damage to the skin. This is further irritated by bacteria, yeasts (eg candida or thrush), detergents, urine (wee) and faeces (poo).

Nappy rash is made worse by:

- runny, loose faeces (diarrhoea)
- not changing the nappy often enough
- plastic pants

Most children grow out of nappy rash with toilet training.

Signs and symptoms

- The skin in the nappy area generally looks red and raw and can spotty in appearance, particularly at the edges of the rash
- It can be sore or itchy when the area is wiped
- Your baby may be unsettled or irritable

2. In assessing Olivia, what are the key components of your history and examination and why?

- Social History (support mechanisms, post natal depression, financial stressors etc)
- Family History
- Immunisations

3. You note that Olivia, is alert, active and thriving, in assessing the nappy rash you note the clinical features are consistent with irritant dermatitis. What clinical features support this diagnosis? How would the clinical findings differ in candidiasis, seborrhoeic dermatitis, miliaria and psoriasis?

- *Seborrhoeic dermatitis*. Presents as a dull red rash covering most of the napkin area, sometimes with a greasy yellow scale, although this is characteristically absent in this moist area
- *Monilia*. Manifested as a thick, white material deep in the folds and as small annular lesions with an overhanging white macerated scale at their margin
- *Irritant dermatitis* from urine and faeces. Irritant dermatitis due to urine affects mainly the convex surfaces, with relative sparing of the flexures; irritation due to faeces particularly affects the natal cleft. Napkin dermatitis is rarely caused by irritant or allergic reactions to laundering products
- *Miliaria*. Sweat duct occlusion may occur alone or in combination with other elements. This presents as small red papules, which are very transient so the pattern varies considerably from hour to hour

- *Psoriasis*. Produces a bright red, glazed, clearly margined napkin rash. This may develop into the condition called 'napkin psoriasis'. A few small scaly spots occur on the trunk above the psoriatic napkin rash, followed by a sudden explosion of typical psoriatic lesions on scalp, face and all over the trunk. The infant is well and the condition is usually asymptomatic. The eruption is self-limiting in a few weeks.

4. What management plan would you recommend to Olivia's mother?

A good quality disposable nappy is best. These allow the moisture to be absorbed quickly, keeping the skin dry. While cloth nappies are good for the environment, they do not take up moisture as well as disposable nappies.

Nappies should be changed often (about 5-7 times a day in babies under 12 months), reducing the contact of urine and faeces with the skin.

At each nappy change your baby's bottom should be gently wiped with cotton wool or 'Chux' type cloths, dampened with luke-warm water. Baby wipes can be quite irritating and should not be used.

A barrier cream should be thickly applied at each nappy change. This will prevent the moisture and irritants from reaching the skin. Zinc cream is the best or white soft paraffin. If the cream wipes off too easily, try another brand because the idea is to create a barrier. These creams are available from your pharmacy, or may be prescribed by your doctor.

Try to let your child have as much time without the nappy on as possible.

Talcum powder should not be used on nappy rash.

Many rashes can occur in the nappy area and may not be due to nappies. Some conditions (such as eczema or a skin infection) which can be found on any area of the body, do not respond to the treatment for nappy rash. A mixture you can make yourself putting 10% of olive oil in zinc paste .

In persistent nappy rash that does not respond to nappy creams, a medicated cream such as an antifungal (eg Canesten, Daktarin, Nystatin) or hydrocortisone (eg Sigmacort 1%) may be necessary. Do not use any stronger cortisone creams in this area unless your doctor tells you to

Antiseptics should not be used on nappy rash

CASE FOUR

Short case number: 4_26_04

Category: Children & Young People; Musculoskeletal system & Skin

Discipline: Paediatrics_Medicine_Dermatology

Setting: General Practice

Topic: Paediatric infectious skin rashes_scabies

CASE



Mitchell Fraser is 4 years old, he presents with his father John who explains that Mitchell has developed this itchy rash on his hands. He has been well, with no history of recent upper respiratory tract infection. He has had no fever. The rash is only on his hands and there are no lesions in his mouth.

His sister has developed a similar rash and his father is concerned that it is something contagious.

QUESTIONS

1. On initial assessment your provisional diagnosis is scabies. What are the key clinical features that would support a diagnosis of scabies?
2. Differential diagnoses include coxsackie B [hand, foot & mouth], impetigo or herpes simplex. What are the key clinical features of these infectious rashes in children?
3. In a table summarise the following impetigo, mollusca contagiosa, tinea, scabies, coxsackie B [hand, foot & mouth], herpes simplex and warts under the headings, causative microorganism, typical presentation, natural history,
4. In relation to the above conditions outline the management and requirements for exclusion from school.

Resources:

1. **Rogers M. Skin disorders in infancy and childhood, Chapter 21 in Practical Paediatrics. Edition 6. Edited by Robertson DM & South M. Churchill Livingstone, Elsevier. [pages 766 - 773]**

ANSWERS

- 1. On initial assessment your provisional diagnosis is scabies. What are the key clinical features that would support a diagnosis of scabies?**

Scabies is caused by a small, eight legged mite called *Sarcoptes scabiei*. Scabies is spread by close physical contact and is common in school-age children. If left untreated, it will usually spread to all members of a child's family. The mites and their eggs may live on clothes or bed linen for 1-2 days.

It is often difficult to diagnose scabies because the symptoms vary and the signs can be very subtle. Commonly, the symptoms include

- Bullae - raised, clear fluid-filled spot or lesion that are bigger than 1cm in diameter
- Papules - solid, raised areas of skin with distinct borders less than 1cm in diameter
- Pustules - raised, fluid-filled spot that appear yellow
- Vesicles - raised, clear fluid-filled spots

Lesions or spots are areas of skin disease and are generally small.

A number of mites burrow into the skin, particularly between the fingers, around the wrists and elbows, the armpit, nipples and penis. In infants and young children, there are often vesicles and pustules on the palms and soles and sometimes on the scalp. It is the allergic reaction to these mites that causes the rash and symptoms.

Sometimes the only symptom is an itch, without a rash. Usually the itch is worse at night and after a hot bath or shower. Bathe your child in warm water as hot water may make the itching worse.

- 2. Differential diagnoses include coxsackie B [hand, foot & mouth], impetigo or herpes simplex. What are the key clinical features of these infectious rashes in children?**

And

- 3. In a table summarise the following impetigo, mollusca contagiosa, tinea, scabies, coxsackie B [hand, foot & mouth], herpes simplex and warts under the headings, causative microorganism, typical presentation, natural history,**

And

- 4. In relation to the above conditions outline the management and requirements for exclusion from school.**

Hand foot and Mouth

Hand foot and mouth disease (HFMD) is a viral infection. It is not linked to the foot and mouth disease that affect animals. HFMD is mainly seen in children under the age of ten or in young adults. It is easily spread from one person to another. Most people have had HFMD by the time they are adults. It rarely causes further complications and is very rarely fatal.

Symptoms usually start three to seven days after catching the infection and can last between seven to 10 days. Admission to hospital is rarely needed. The common signs and symptoms include:

- high temperature (fever)
- sore throat
- small blisters on the inside of the mouth, the sides of the tongue, palms of the hands, fingers, soles of the feet and nappy area (the blisters should not be itchy like chickenpox blisters).
- poor appetite
- drinking and eating can be painful because of the mouth blisters
- tiredness

The main way HFMD is spread is by touching the fluid from inside the blisters, and fluids from the nose, mouth and chest (spread from sneezing and coughing). It can also be in bowel movements (poo) for up to several weeks after being infected.

How to stop it spreading?

- Washing hands after touching these bodily fluids is the best way to prevent the spread of HFMD.
- Not sharing items like cutlery, drinking cups, towels, toothbrushes and clothing will help to reduce the spread to others.
- You/your child should stay home from school, crèche, play group, kindergarten, or child care until all the fluid in the blisters has dried.

Impetigo

Impetigo is an infection of the skin caused by bacteria (usually by one of two bacteria called either staphylococcus aureus or group A, b haemolytic streptococcus). These bacteria can live on the skin, in the throat or nose, or on other parts of the body without causing a problem but sometimes they start to cause an infection such as impetigo.

- The sores can be anywhere on the body, but are often on the face near the mouth and nose, or on the arms and legs.
- Impetigo may start with a blister or a group of blisters.
- The blister bursts leaving a patch of red, wet skin which weeps.
- The spot usually becomes coated with a tan or yellowish crust, making it look like it has been covered with honey.
- There can be small spots around the first spots, spreading outwards.
- Impetigo is usually itchy.

- The sore takes about 1 to 3 days to develop after contact with fluid or crusts from a sore.
- There is often superficial peeling on the edge.

Herpes Simplex

Herpes simplex gingivostomatitis is inflammation of the gums and lips caused by the herpes virus. This virus causes cold sores.

In children go unnoticed but approximately one child in four will develop mouth blisters after their first infection. Your child may have a fever, become irritable, and a day or two later develop ulcers on the lips and gums. These are painful and often lead to drooling and refusal to eat or drink. Your child may then get dehydrated. The blisters usually take between 10–14 days to go away. The blisters never leave scars.

Signs and symptoms

- fever
- irritability
- mouth ulcers/blisters
- pain
- poor appetite
- dehydration

The infection can be passed on to other children if they come into contact with fluid from the blisters. Children should stay home from school until all blisters are healed and child is well.

Warts

Warts are generally harmless skin growths. They are very common, affecting around one in 10 children. Warts can come up anywhere on the skin or mucous membranes (eg mouth, genitals) however, most commonly on the hands, fingers, feet and face.

Warts can spread, usually due to direct contact with someone with warts. Picking or scratching may lead to spread of warts on the same person. You can also get them indirectly through swimming pools or public showers especially if barefoot and you have scratches or cuts.

Half of all warts will disappear within 2 years, but this is not predictable. They can last longer in adults, or people with immune system problems.

There are many different types of wart virus, and you may become immune to one type, but develop warts due to another type. The immunity is not lifelong.

- warts can have many different appearances, from flat, shiny, long stalks, or large rough lumps
- they can be unsightly and embarrassing
- they can be painful, particularly when they are on the feet or under nails

- Warts are caused by a family of viruses called human papilloma virus (HPV). Once infected with the virus, it may take up to a year or more for the wart to become visible. They then grow very slowly over a period of months to years.
- Children do not have to stay home from school however it is recommended the warts be covered with a bandage.

Molluscum

Molluscum is a common skin problem with small, harmless raised spots. The spots can stay on the body for a few weeks to several months or more than a year. It is caused by the Molluscum Contagiosum Virus (MCV). This virus only lives in humans.

Molluscum is contagious and can be spread from person to person or to different areas in the same person. It can also be spread by children sharing baths or pools through bath toys and towels. Usually it does not need any treatment. Good personal hygiene can help prevent it spreading.

At first the molluscum spots look like pimples. Then they become round, pearl coloured lumps that have a white mark or core in the centre of them. They are usually between 1 to 5 mm in size, but can be as big as 1 - 2 cm wide. Usually they are on the nappy area, tummy, face, arms and legs.

Molluscum spots usually heal without scarring, although a small 'chicken pox' type scar is possible. Scarring is slightly more likely with more aggressive treatment. Molluscum spots are painless but can be itchy - scratching can also cause scarring.

Molluscum is spread from skin to skin contact when touching the molluscum lesions/lumps. Sharing towels and face cloths is another common way to spread the virus. Swimming in pools can also spread it to other children through the pool water. It can take weeks to months from when your child has contact with the molluscum virus to when the spots appear. There is no need to have time off school however lesions should be taped and covered.

CASE FIVE

Short case number: 4_26_05

Category: Children & Young People; Musculoskeletal system & skin

Discipline: Paediatrics_Medicine_Dermatology

Setting: Paediatric Outpatients.

Topic: Vascular Birthmarks_haemangiomas.

CASE



Jade Spader is 6 months old; she has been referred to the paediatric outpatient clinic by her General Practitioner, because of the rapidly growing lesion on her forehead.

Jade was born by normal vaginal delivery at term following an uncomplicated pregnancy. She is the first child of Aaron and Jennifer. There was no lesion noted at birth, by 4 weeks a smaller pink spot was noticed on Jade's forehead, it has continued to rapidly grow in size since then and her parents are quite worried.

Jade is otherwise well, her growth and development have been normal

QUESTIONS

1. What are the key features of history and examination that assist in differentiation between haemangiomas and vascular malformations?
2. Your diagnosis is that Jade has a superficial haemangioma. What you explain to Jade's parents about the underlying pathophysiology of this lesion and the natural history?
3. Outline the possible complications of haemangiomas.
4. Outline your management plan for Jade?

Resources:

1. Rogers M. Skin disorders in infancy and childhood, Chapter 21 in Practical Paediatrics. Edition 6. Edited by Robertson DM & South M. Churchill Livingstone, Elsevier. [pages 764 - 765]

1. What are the key features of history and examination that assist in differentiation between haemangiomas and vascular malformations?

Vascular birthmarks can be divided into:

- haemangiomas, which are proliferative vascular tumours
- vascular malformations, which represent fixed collections of dilated abnormal vessels.

Haemangiomas are common birthmarks which are usually red or purple. They mostly occur on the head and neck areas and develop shortly after birth.

Haemangiomas appear in the first weeks of life. They are often confused with a scratch or bruising related to birth trauma.

They then become a flat red patch often with blood vessels you can see. If they are on the skin surface they can look like a strawberry. If they are deeper in the skin they look like a bluish lump. Sometimes they have both a surface and a deep part. Haemangiomas can also grow on the internal organs. These behave in the same way as those on the skin.

Haemangiomas grow fast in the first few months then slow down in the second six months of life. Sometimes they keep growing after the child's first birthday. When they have stopped growing, they tend to stay the same size for a while then start to disappear. Some will go away completely while other will leave behind a soft fatty lump. It is variable how quickly they disappear. Some have nearly gone by 3 years, while a small number (less than 1 in 10) are still there up to 8 or 9 years old

Vascular malformations are collections of dilated abnormal vessels divided according to the vessels of origin. All vascular malformations are present at birth and grow only in proportion to the growth of the child. They show no tendency to involution.

2. Your diagnosis is that Jade has a superficial haemangioma. What you explain to Jade's parents about the underlying pathophysiology of this lesion and the natural history?

Haemangiomas are caused by an overgrowth of the cells that line blood vessels (arteries and veins). This over growth of cells forms lesions or birthmarks. Most haemangiomas disappear by themselves and do not need any treatment.

3. Outline the possible complications of haemangiomas.

- Incomplete resolution - redundant tissue, residual telangiectasia, scarring following ulceration
- Ulceration - full-thickness tissue loss on 'edge structures' (lip, lid, ala), inevitable scarring, cicatricial ectropion from scarring of eyelids, cicatricial alopecia

- Obstruction - of eye, producing amblyopia, of nose, leading to difficulty in breathing during feeding, of lip, leading to problems with sucking, of larynx (a 'beard' distribution haemangioma is a marker for possible laryngeal involvement).

4. Outline your management plan for Jade?

Haemangiomas need treatment if:

- They are pressing on important structures such as the eyes, nose, ears or windpipe
- It is on your child's eyelid or if it is large and on your child's face. These haemangiomas need urgent treatment to prevent serious problems.
- They become ulcerated and painful and do not heal
- To prevent your child having problems with their appearance when they reach school age

Treatment should be given when the benefits of treatment outweigh the risks involved. All treatments have side effects which your doctor will discuss with you.

Cortisone

Cortisone can be taken by mouth or injected directly into the haemangioma. If this does not work, other medicines such as *Propanolol*, *Vincristine* and *Interferon* may be used.

Laser therapy

Laser treatment can sometimes help treat a haemangioma, especially if used very early in life while the haemangioma is still flat. Laser is a high energy light source. The particular laser used, *Candela V-beam pulsed dye laser*, only targets blood vessels. (See fact sheet: Dermatological (skin) laser therapy).

This means it destroys the blood vessels that make up the birthmark without injuring the skin over the top. An anaesthetic and many treatments may be needed.

Surgery

In a few cases, haemangiomas are removed by surgery during the first year of life. Usually surgery is delayed for several years until the haemangioma has shrunk a lot.

When your child's haemangioma has mostly disappeared, there may still be some scarring, loose skin or fatty lumps. These might need surgery or laser.

Care at home

- Haemangiomas do not usually need any special care
- If the haemangioma bleeds, put firm pressure on the area for 10 minutes, or until the bleeding stops. Tell your doctor.
- If an ulcer develops it can be painful and your child may be unsettled or distressed. You should visit your doctor who may prescribe a local anaesthetic cream to help reduce the pain and/or antibiotic cream for infection. Once you have put the cream on the ulcer, you will need to cover the ulcer with a dressing. Haemangiomas that are ulcerated can be quite slow to heal.